

GENERAL RISK ASSESSMENT TEMPLATE



Work area / operation	CB11.10.403	Assessor's name	JACK HEFFERNAN, JAMES WITCOMB, WILL RICHARDSON
Other persons consulted	Zain Khan	Date of safety assessment	
Subject Coordinator's Name	Gavin Paul	Lab Supervisor's Name	Felipe Ulloa Gutierrez

ACTIVITY - Describe hazardous activities related to the work area or operation.	ASSOCIATED HAZARDS	INHERENT RISK - Harm that could occur from these hazards if controls fail or are not in place.	EXISTING CONTROL MEASURES	PROPOSED CONTROL MEASURES - Proposed action to minimise risk to an acceptable level.	TARGET DATE - To implement proposed controls	RESIDUAL RISK LEVEL (H,M,L)
Electrical cables mishandled	Tripping, electrical shock, cable damage	Trip and falling, electric shock.	Cables routed in safe ways, insulated wiring.	Cable management, Visual inspection, no cables across walkways	Before lab sessions (ASAP)	L
Leaving the robot arm unattended	Uncontrolled and unexpected movement, collision with bystanding.	Unexpected arm movement hitting people, collision with equipment.	E stop, supervision from tutors.	Don't leave the robot unattended, power down and utilise e stop before leaving.	Before lab sessions (ASAP)	L
Using inappropriate load on the robot	Mechanical failure, arm overload, dropping payload.	Mechanical failure or joint damage.	Manager payload specifications.	Check payload ratings before using robot.	Before lab sessions (ASAP)	L
Robot arm moves incorrectly	Collision with environment, unexpected trajectories	Collision with operator or nearby equipment.	Simulation settings, before the robot is run, E stop.	Simulate robot before using physically, low speed run before normal run speed.	Before lab sessions (ASAP)	L
Incorrect manual handling	Injury to robot from lifting robot in unintended ways	Injury to robot from moving the robot incorrectly.	Lab equipment being in a fixed position usually.	Manual handling training guidance from tutors, use trolley or wheels when moving robot.	Before lab sessions (ASAP)	L
Operating robot when tired or distracted	Human error from incorrect commands	Operator or environment colliding with robot.	Scheduled supervision from markers.	Don't use the robot when tired, use a buddy when using the robot.	Before lab sessions (ASAP)	L
Programming while robot is active	Accidental motion leading to unintended movements from robot.	Operator being hit and moving in other unintended ways within workplace.	Reduce speed mode, Supervision, E stop.	Stay outside of robot range whilst coding, upload code whilst machine is powered down or in emergency stop mode.	Before lab sessions (ASAP)	L

Pick-and-place of bar props	Gripper drops or flings item mid-move, pinch points at gripper fingers	Operator struck by object, pinched fingers, item broken	E stop, reduced speed mode	Plastic props only (no glass), stay under payload, test grip at low speed first, keep hands clear of gripper while powered	Before lab sessions (ASAP)	L
E-stop / PLC wiring	Incorrect wiring means E stop doesn't cut robot motion	Robot keeps moving during an emergency, operator hit	Supervision from tutors	Test E stop and reset/recovery before every run, supervisor checks wiring before powering on	Before lab sessions (ASAP)	L
Running sim code on the real robot	Real base/table positions differ from simulation, unexpected trajectories	Arm collides with table, props or operator	Simulation, E stop	Measure real setup and match it in sim, jog to first pose manually at low speed, dry run with no props first	Before lab sessions (ASAP)	L
Working near the moving arm (entanglement)	Loose clothing, long hair or lanyards caught by arm or gripper	Entanglement, being pulled into robot	Lab safety brief	Tie back long hair, remove lanyards and loose clothing, no gloves near moving arm, stay outside robot range while running	Before lab sessions (ASAP)	L
Liquids near equipment	Spill onto controller/PLC, wet floor	Electric shock, slipping	Spill kit in lab	Electrics kept away from liquids, paper towels on hand, clean spills immediately	Before lab sessions (ASAP)	L

Approval of assessment	I am satisfied that the residual risk with existing controls is acceptable ☒ Yes ☐ No	Signature	JACK HEFFERNAN, JAMES WITCOMB, WILL RICHARDSON	Date	21/08/2026
	OR I am satisfied that that the proposed controls will reduce risk to an acceptable level. ☒ Yes ☐ No				

Guidance notes for documenting General Risk Assessments

ACTIVITY

Briefly describe this hazardous work activity - E.g. Operating, Handling, Using ... (Include names) of hazardous equipment, substances or materials used, and any quantities and concentrations of substance(s) or reaction products.

ASSOCIATED HAZARDS

Plant & Equipment – noise, vibration, moving parts (crushing, friction, stab, cut, shear), pressure vessels, lifts/hoists/cranes, sharps

Manual Handling – repetitive movements, lifting awkwardly, lifting heavy objects

Work Environment – moving objects, extremes in temperature, isolation, work at height, allergies to animal bedding, dander and fluids, risk of fire/explosion, slippery surfaces/trip hazards

People – potentially violent or volatile clients/interviewees

Communicable Diseases – exposure to bodily fluids/infectious materials, animal bites and scratches,

Environmental – emissions to atmosphere, discharge to soil and water bodies (including stormwater run-off), nuisance noise & odour, poor ventilation/air quality

Radiation (non-ionizing) – including lasers, microwaves or UV light

Electrical – plug-in equipment used in 'hostile' work environment, exposed conductors, high voltage equipment

Pathogens – dealings with pathogenic microorganisms such as bacteria, parasites, fungi or viruses

GMOs – dealings with genetically modified organisms

Cytotoxins – carcinogens, mutagens or teratogens

Radiation (ionizing) – ionizing radiation source such as radioactive substance or radionuclide, or irradiating apparatus

Chemical – hazardous substances, dangerous goods, fumes, dust, compressed gas, hazardous waste

INHERENT RISK

Provide details of the harm that could be caused to people or the environment if something goes wrong.

For example: inhalation of fumes, laceration, injury to back, infection, burns to skin or eyes.

Think about what could happen if controls fail or are not in place.

CONTROL MEASURES

Note the existing and proposed actions to reduce risk to an acceptable level. Apply the "Hierarchy of Controls", listed below, when deciding the best control measure to apply. Control types closer the top of the list are preferable.

1. **ELIMINATE THE HAZARD.** For example: use a different less dangerous piece of equipment, fix faulty machinery, use safer materials or chemicals
2. **ISOLATE THE HAZARD FROM THE PEOPLE.** Separate people from the danger. For example: use shielding, use lifting equipment or trolleys, remove dust or fumes with exhaust system, lock-out machinery.
3. **CHANGE THE WAY THE JOB IS DONE.** For example: change work practices, provide training, information and signs, develop work procedures.
4. **USE PERSONAL PROTECTIVE EQUIPMENT (PPE),** noting specific PPE is required for each job. For example: respirator, hearing protection, gloves. Training and information is required for the use of PPE.

RESIDUAL RISK LEVEL (H, M, L)

Estimate risk taking into account the way the activity is run and control measures put in place. The level of risk can be determined by combining consequence and likelihood using the risk matrix from below. Residual risk should be reduced to a level acceptable by management.

CONSEQUENCE OF HARM - This is how bad it will be if something does go wrong e.g. the number of people that could be harmed, the severity of injury.

LIKELIHOOD OF HARM - Chance of harm occurring is affected by the duration of the activity and its frequency; the number of people doing the activity and the level of exposure to the hazard.

		CONSEQUENCE					
		Insignificant	Minor	Moderate	Major	Catastrophic	
		Injury/illness consequence	Non-injury incident	Injury/ill health requiring first aid	Injury/ill health medical attention	Injury/ill health requiring hospital admission	Fatality or permanent disabling injury
		Environmental consequence	Minor effects on biological or physical environment	Moderate short term effects but not effecting ecosystem functions	Serious medium-term environmental effects	Very serious long term impairment of ecosystem functions	
LIKELIHOOD	Almost Certain	The event will occur on an annual basis	Moderate	High	High	Critical	Critical
	Likely	The event has occurred several times or more in your career	Moderate	Moderate	High	High	Critical
	Possible	The event might occur once in your career	Low	Moderate	Moderate	High	High
	Unlikely	The event does occur somewhere from time to time	Low	Low	Moderate	Moderate	High
	Rare	Heard of something like this occurring somewhere	Low	Low	Low	Moderate	Moderate

UTS SAFE WORK METHOD STATEMENT (SWMS)

1. FACULTY/SUBJECT			
Faculty/Subject title		41013 Industrial Robotics	
Subject supervisor/coordinator		Gavin Paul	
SWMS prepared by		Jack Heffernan	

2. WORK ACTIVITY DESCRIPTION	
Describe the work activity E.g. Operating, Handling, Using.. Include names of hazardous equipment, substances or materials used, and any quantities and concentrations of substance(s) or reaction products.	Programming and operating a UR3e collaborative robot arm in the robotics lab to perform pick-and-place tasks for a robotic bartending demonstration. Motions are simulated in Python (Swift) first, including collision and joint limit checks, before a basic movement of the UR3e is run on the physical robot at reduced speed. In terms of the virtual environment, hazards such as robot-to-robot collisions, collisions with the bar and objects, and a person entering the workcell are modelled and addressed in simulation using barriers, a light curtain, an E-stop and collision avoidance. Spillage and other bar factors are also taken into account. The activity includes connecting and handling electrical connectors (UR controller, PLC and E-stop circuit), attaching the gripper end effector, and manual handling of equipment and props.

3. HAZARDS: Choose those hazard types that will need to have control measures in Section 4			
Work Environment - Slip/Trip/Fall hazards - Water/alcohol spills	Plant - Noise - Vibration - Moving parts (Crushing, friction, cut, stab, shear hazards) - Dropped or flung objects - Collision within workspace - Unexpected motion - Parts being entangled with the robot	Chemical	Ergonomic/Manual Handling - Working above shoulder or below knee height - Poor workstation set up - Repetitive or awkward movement
Electrical - Plug in equipment - High voltage - Exposed wiring - Exposed conductors	Radiation	Biological	Psychosocial - Working in isolation - Working with timeframes

4. CONTROLS MEASURES: Choose those that apply for hazards identified		
Eliminate/Isolate/Substitute / Engineering Controls - Remove hazard - Restrict access - Regular maintenance of equipment - Redesign of workspace / workflow - Use plastic props instead of glass with liquid to stop shattering or spills in case of malfunction. - Reduce speed. - Simulate software joint limits.	Admin specific: Licenses/permits Work Methods - Training Information or Instruction - Licensing or certification of operators - Test and tag electrical equipment - Regular breaks - Task rotation - Work in pairs - Have pre session checks of each component	Emergency Response Systems - First aid kit - Emergency Stop button - Remote Communication Mechanism
Other controls not listed Tie back all loose hair and clothing.		

5. PPE REQUIRED (Tick those that apply)
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 EYE PROTECTION	 HEARING PROTECTION	 CLOSED-IN FOOTWEAR MUST BE WORN	 HAND PROTECTION	 FACE SHIELD	 PROTECTIVE CLOTHING
☒	☐	☒	☐	☐	☐
 RESPIRATORY PROTECTION	 LONG HAIR MUST BE CONTAINED	 HEAD PROTECTION			
☐	☒	☐			

6. EMERGENCY EQUIPMENT

 EMERGENCY SHOWER	 EMERGENCY SPILL KIT INSIDE	 EMERGENCY EYEWASH
☐	☒	☐

7. WORK ACTIVITY STEPS

BEFORE YOU START:

- Check E stop is working
- Closed in footwear is worn.
- Confirm workspace is clear
- Tie back long hair and wear safety glasses.
- Confirm payload is within safe bounds.
- Confirm that the electrical parts are plugged in safely.
- Virtually simulate the robot moving before starting.

STEPS IN WORK ACTIVITY:

1. Power on the robot and confirm E stop is engaged
2. Load and verify the program is correct
3. Ensure Area is cleared and run program at reduced speed.
4. Monitor movement and hold over e stop in case something happens
5. Return robot to original position
6. Pack up, power down and ensure workspace is left clean
7. Log report, recording incidents if they occurred and store equipment safely.

EMERGENCY PROCEDURES:

- Press emergency button
- Notify security or dial 6 using the UTS internal phone

TRAINING REQUIRED:

- Robot operation induction / lab safety brief
- E stop emergent training

8. SIGN OFF

PREPARED BY:

LAB SUPERVISOR

DATE: 24/09/2026

NAME: JACK HEFFERNAN, JAMES
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REVIEW DATE: _____